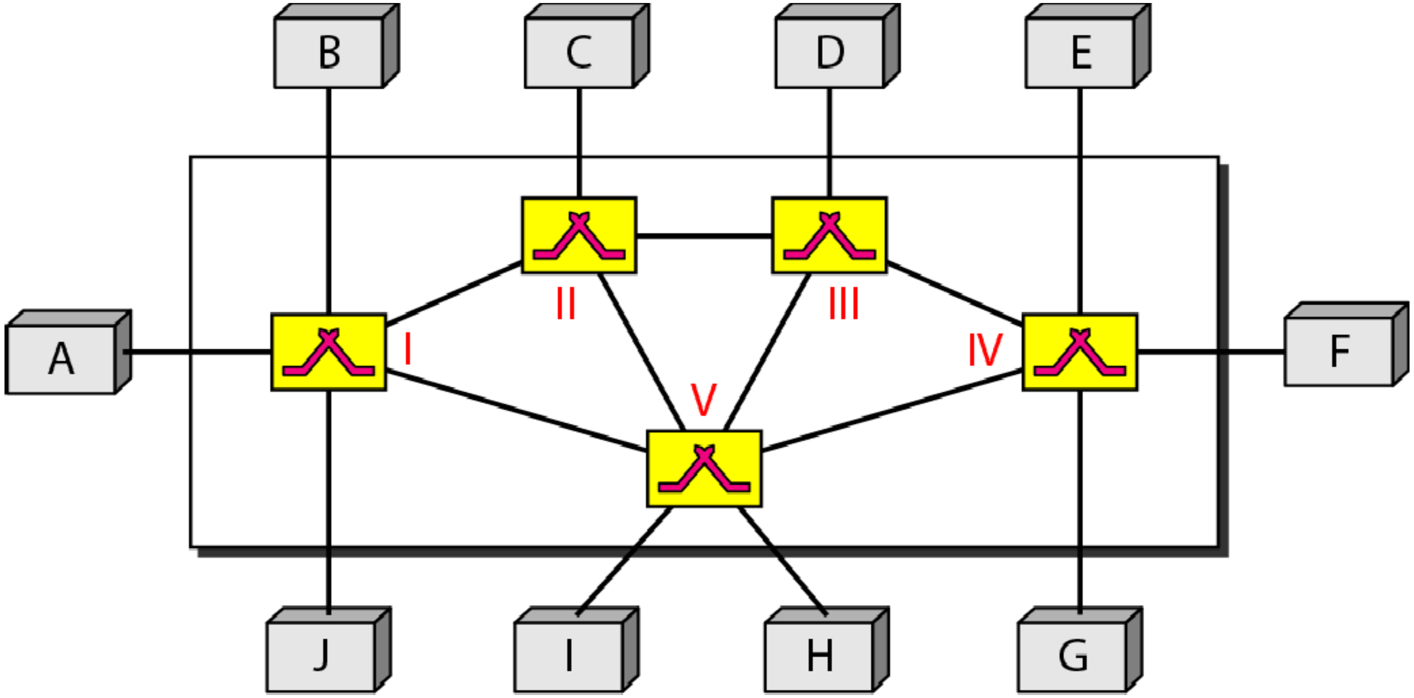
Module: 2 Circuit and Packet Switching

Switched Communication Networks – Circuit Switching – Packet Switching – Comparison – Implementing Network Software, Networking Parameters (Transmission Impairments, Data Rate and Performance)

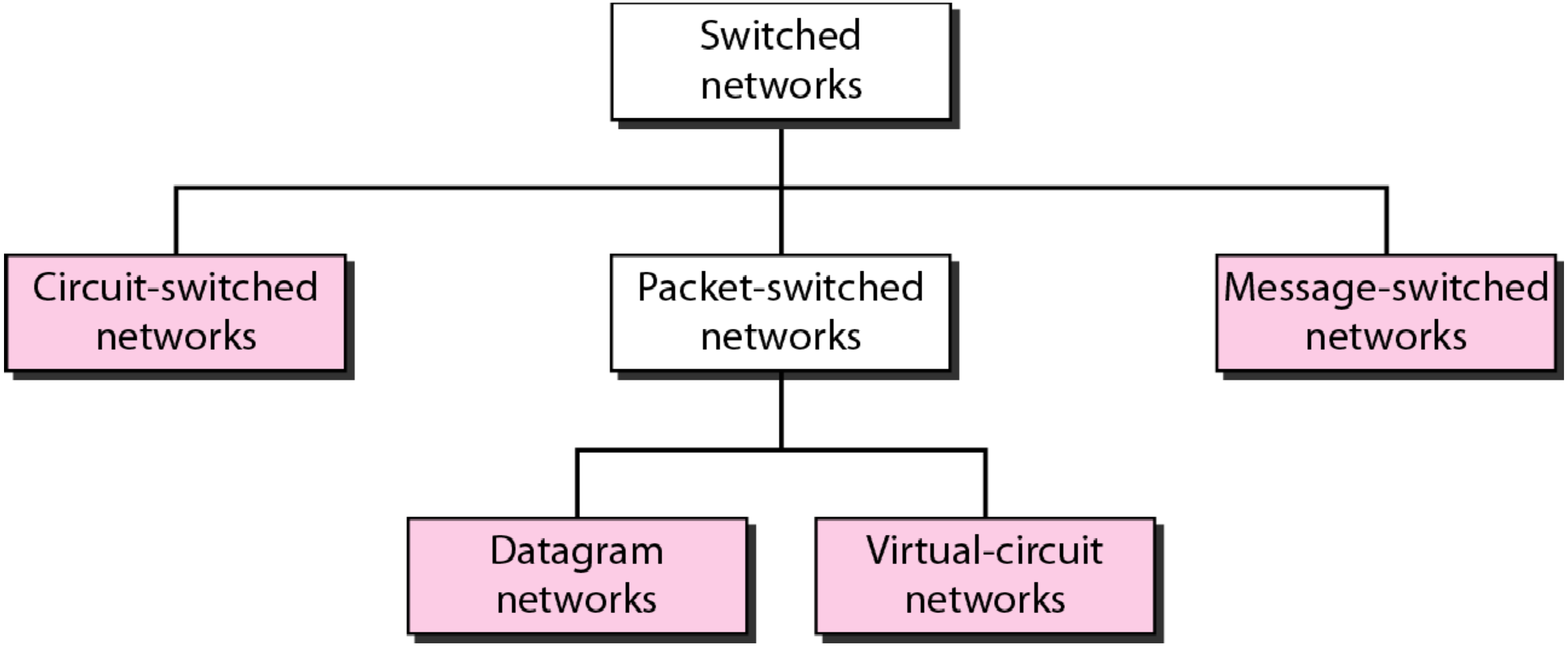
NEED FOR SWITCHING

- Whenever we have many devices, the interconnection between them becomes more difficult as the number of devices increases. Some of the conventional ways of interconnecting devices are:
  - Point-to-point connection between devices as in a **mesh topology**
  - Connection between a central device and every other device – as in **star topology.**
  - **Bus topology** – not practical if the devices are at great distances
- All these **techniques** require **extensive cabling**, dependence on a **central server** or a **central bus.**
- The **better solution** to this interconnectivity problem is ***switching***.

Switched network



Taxonomy of switched networks



CIRCUIT-SWITCHED NETWORKS

- A circuit-switched network consists of a set of switches connected by physical links.
- A connection between two stations is a dedicated path made of one or more links.
- However, each connection uses only one dedicated channel on each link.
- Each link is normally divided into n channels by using FDM or TDM.

Note

A circuit-switched network is made of a set of switches connected by physical links, in which each link is divided into *n* channels.

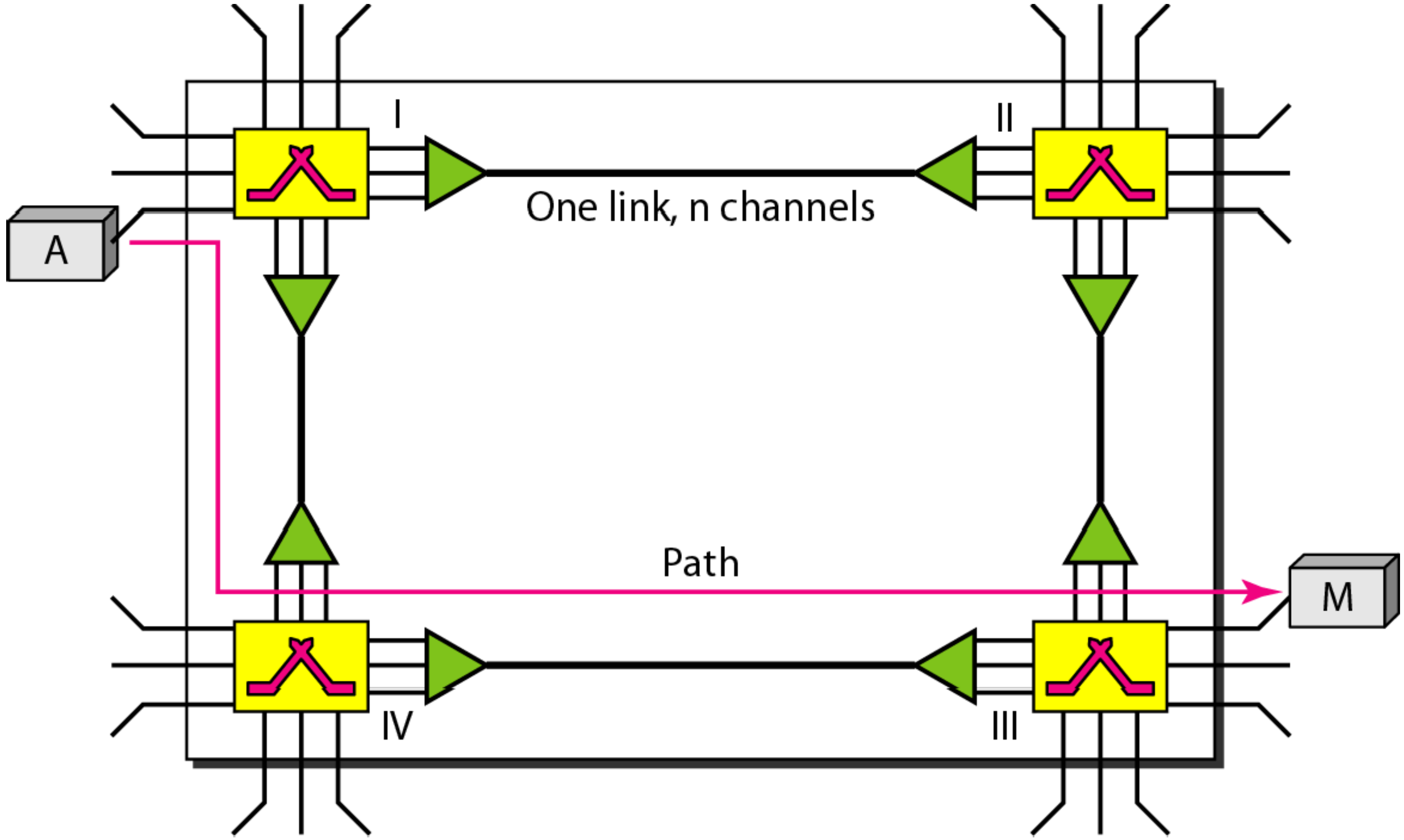
PHASES IN CIRCUIT-SWITCHING NETWORKS

- The actual communication in circuit- switching requires

  3 phases:

  - **connection setup or circuit establishment,**
  - **data transfer, and**
  - **connection teardown or circuit disconnect.**

A trivial circuit-switched network

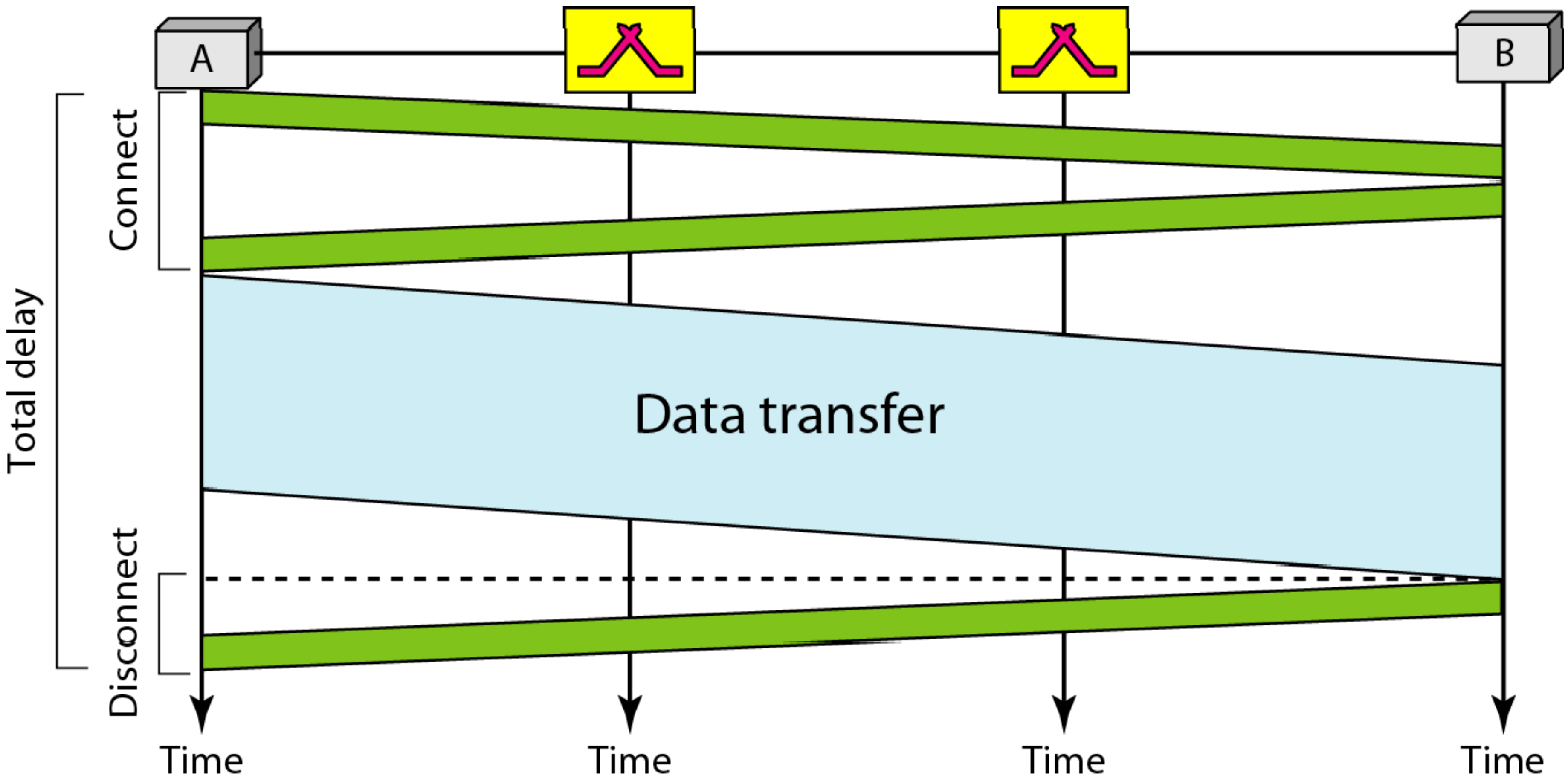


- Before the two systems can communicate, a **dedicated circuit** (i.e. a combination of channels and links) must be established.
- For example, in fig, when **system A** needs to communicate to **system M**, it sends a **setup request** to **switch I**. This request contains the **address** of **system M**.
- **Switch I** finds a channel between itself and **switch IV.**
- **Switch I** then **sends the request** to **switch IV**. Switch IV finds a dedicated channel between itself and **switch III**.
- **Switch III** informs **system M** that system A wants to communicate.
- **System M** then sends an **acknowledgement** in the opposite direction to **system A**. Once **A** receives the **acknowledgment**, the connection is established.

Note

In circuit switching, the resources need to be reserved during the setup phase; the resources remain dedicated for the entire duration of data transfer until the teardown phase.

Delay in a circuit-switched network



A path in a digital circuit-switched network has a data rate of l Mbps. The exchange of 1000 bits is required for the setup and teardown phases. The distance between two parties is 5000 km. Answer the following questions if the propagataion speed is 2×10^8 m:

a. What is the total delay if 1000 bits of data are exchanged during the data transfer phase?

b. What is the total delay if 100,000 bits of data are exchanged during the data transfer phase?

c. What is the total delay if 1,000,000 bits of data are exchanged during the data transfer phase?

Solution:

Given:

Total bandwidth or data rate = 1 Mbps

Distance between 2 parties = 5000 kms

1000 bits are required for setup and teardown phases (Message size)

Propagation speed = 2×10^8 m

Formula:

Total delay = Propagation delay + Transmission delay

Transmission delay = Message size / Data rate

Propagation delay = Distance / Propagation speed

Note

Switching at the physical layer in the traditional telephone network uses the circuit-switching approach.

Packet Switching

- In data communications, we need to send messages from one end system to another.
- If the message is going to pass through a different network, it needs to be divided into packets of fixed or variable size.
- In packet switching, there is no resource allocation for a packet.
- Resources are allocated on demand.
- When a switch receives a packet, no matter what the source or destination is, the packet must wait if there are other packets being processed.
- We can have two types of packet-switched networks: datagram networks and virtual circuit networks.

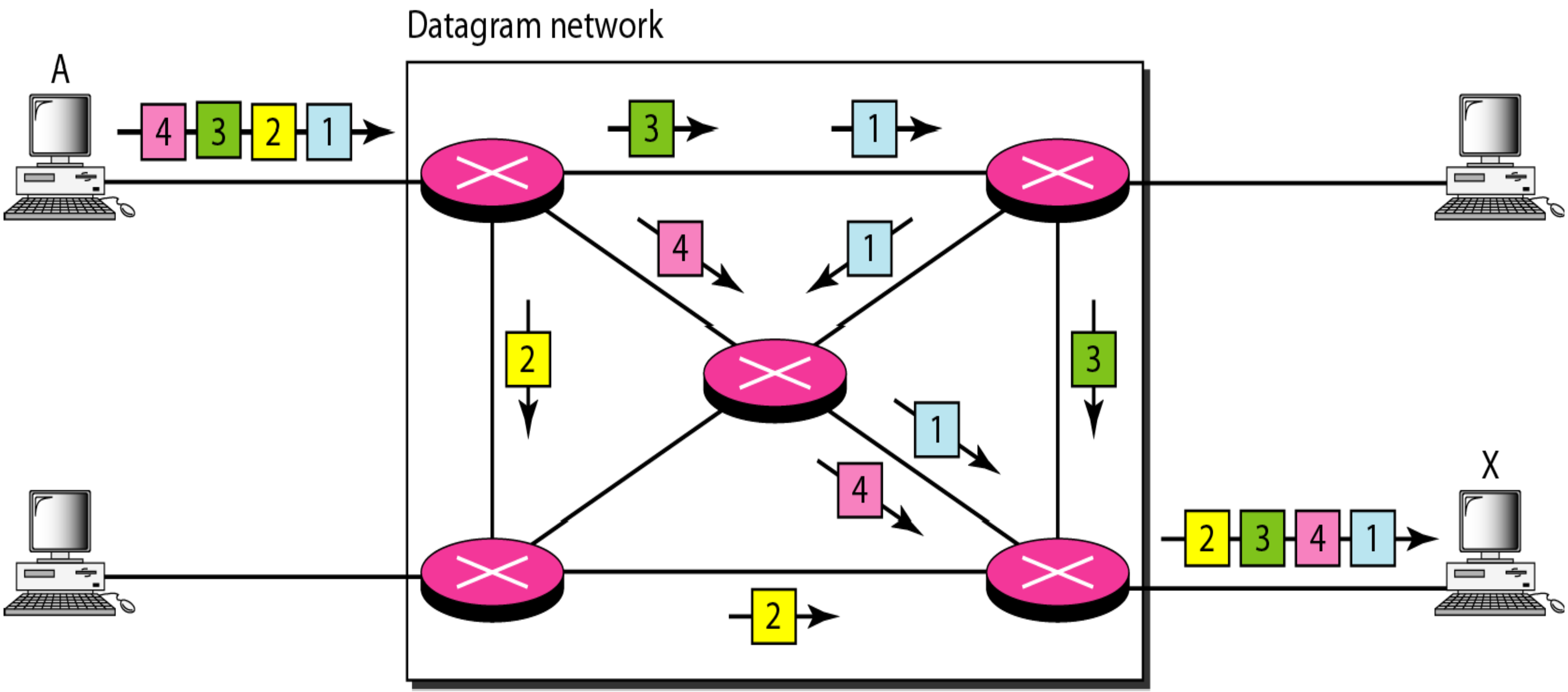
Note

In a packet-switched network, there is no resource reservation; resources are allocated on demand.

Datagram Networks

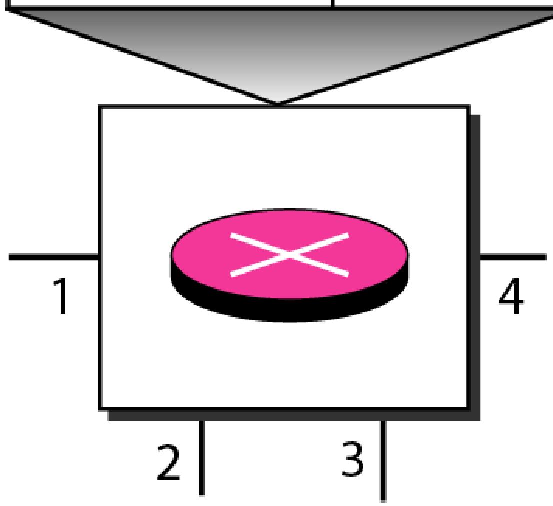
- Each packet is treated independently of all others.
- Even if a packet is part of a multipacket transmission, the network treats it as though it existed alone.
- Packets in this approach are referred to as ***datagrams.***
- Datagram switching is normally done at the network layer.

A datagram network with four switches (routers)



Routing table in a datagram network

Destination address	Output port
1232	1
4150	2
⋮	⋮
9130	3



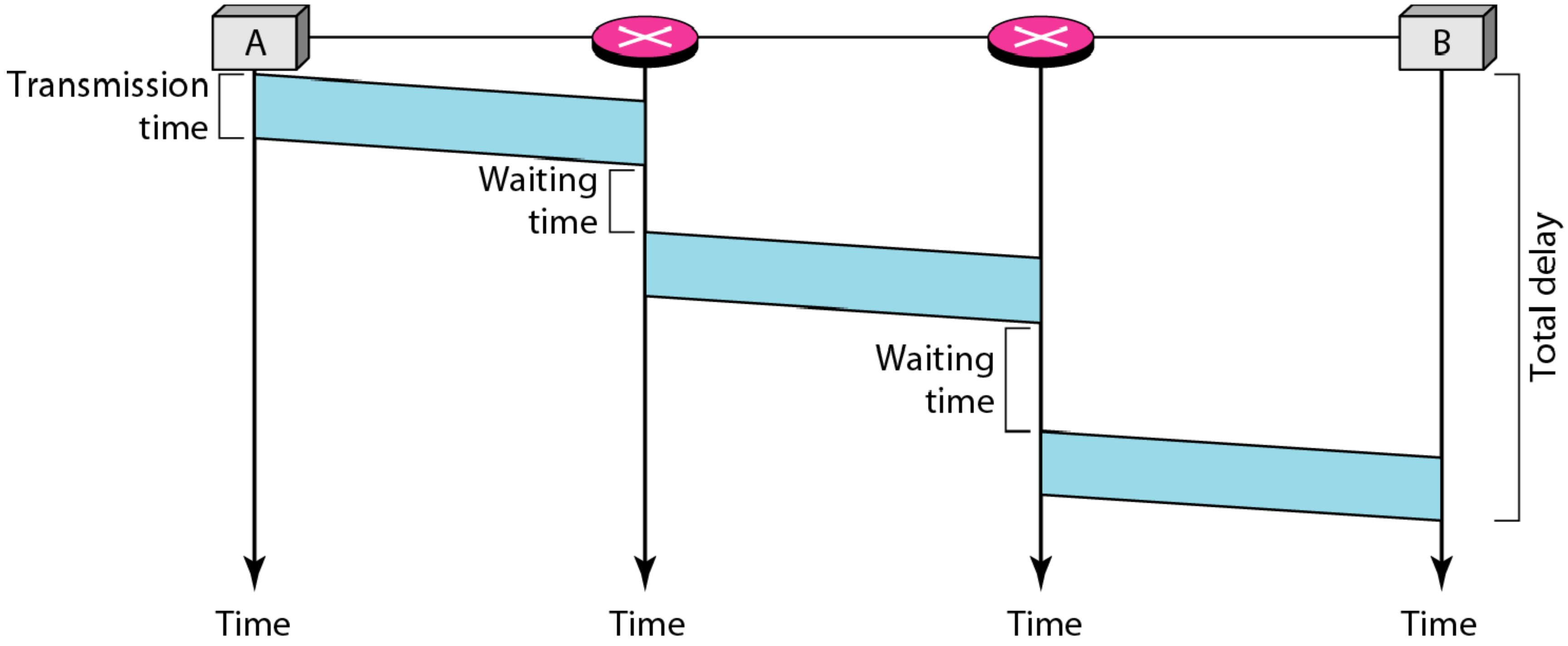
Note

A switch in a datagram network uses a routing table that is based on the destination address.

Note

The destination address in the header of a packet in a datagram network remains the same during the entire journey of the packet.

Delay in a datagram network



Five equal-size datagrams belonging to the same message leave for the destination one after another. However, they travel through different paths as shown in Table 8.1.

Table 8.1 *Exercise 12*

Datagram	*Path Length*	*Visited Switches*
1	3200Km	1,3,5
2	11,700 Km	1,2,5
3	12,200 Km	1,2,3,5
4	10,200 Km	1,4,5
5	10,700 Km	1,4,3,5

We assume that the delay for each switch (including waiting and processing) is 3, 10, 20, 7, and 20 ms respectively. Assuming that the propagation speed is 2 x 10^8 m, find the order the datagrams arrive at the destination and the delay for each. Ignore any other delays in transmission.

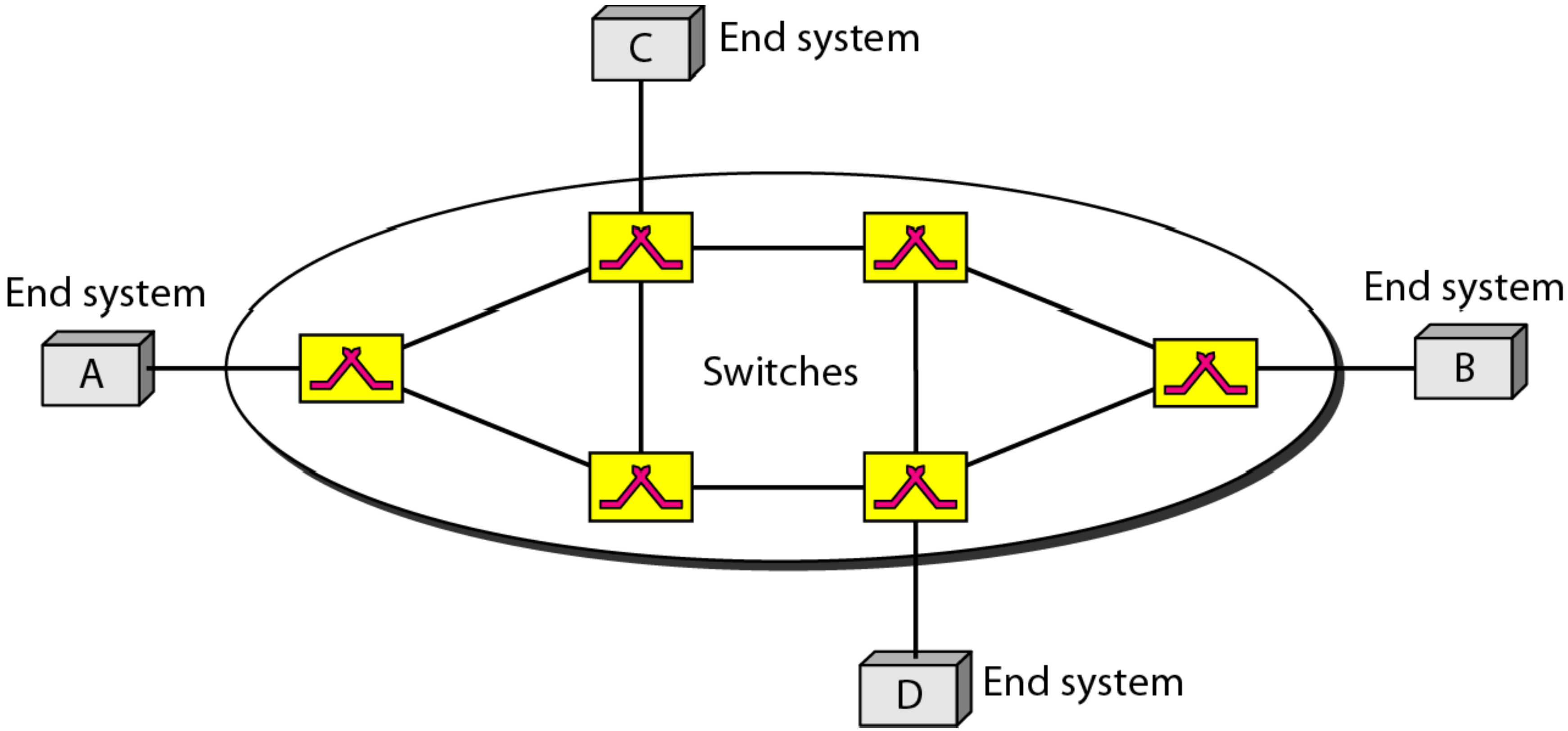
Note

Switching in the Internet is done by using the datagram approach to packet switching at the network layer.

VIRTUAL-CIRCUIT NETWORKS

- A virtual-circuit network is a **cross between a circuit- switched network and a datagram network**.
- It has some characteristics of both.
- As in **CSN**, there are **setup, data transfer and teardown phase**
- Resource can be allocated during the setup phase, as in CSN, or on demand, as in datagram network
- As in datagram network, data are packetized with address
- However, the address is local jurisdiction not End-to-end
- A VCN, is implemented in DLL, while CSN is in PL and datagram network is in N/w layer

Virtual-circuit network

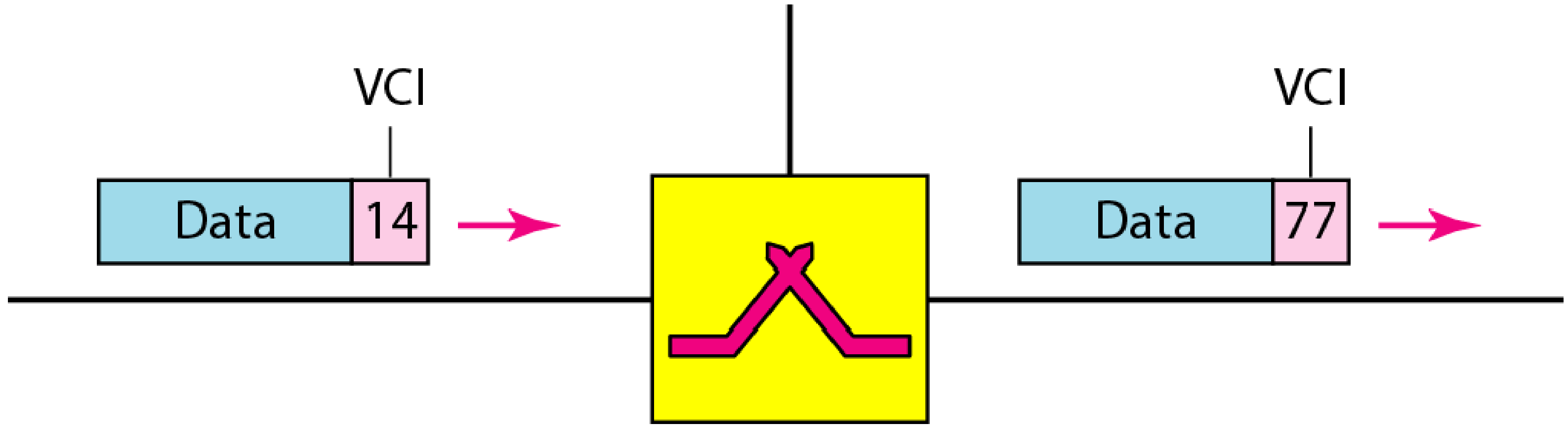


Addressing

Two types of addressing

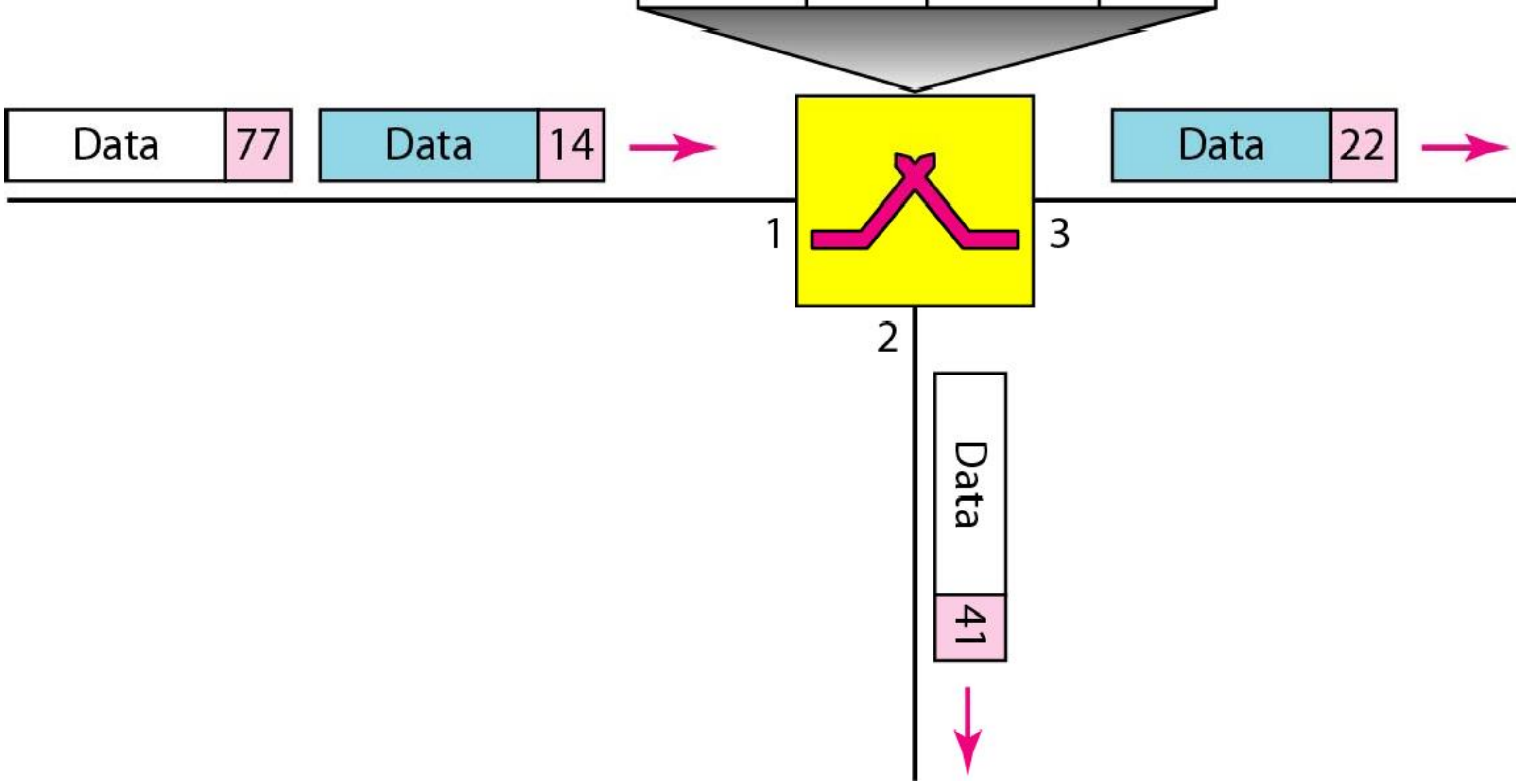
1. Global
2. Local (Virtual – circuit identifier)

Virtual-circuit identifier

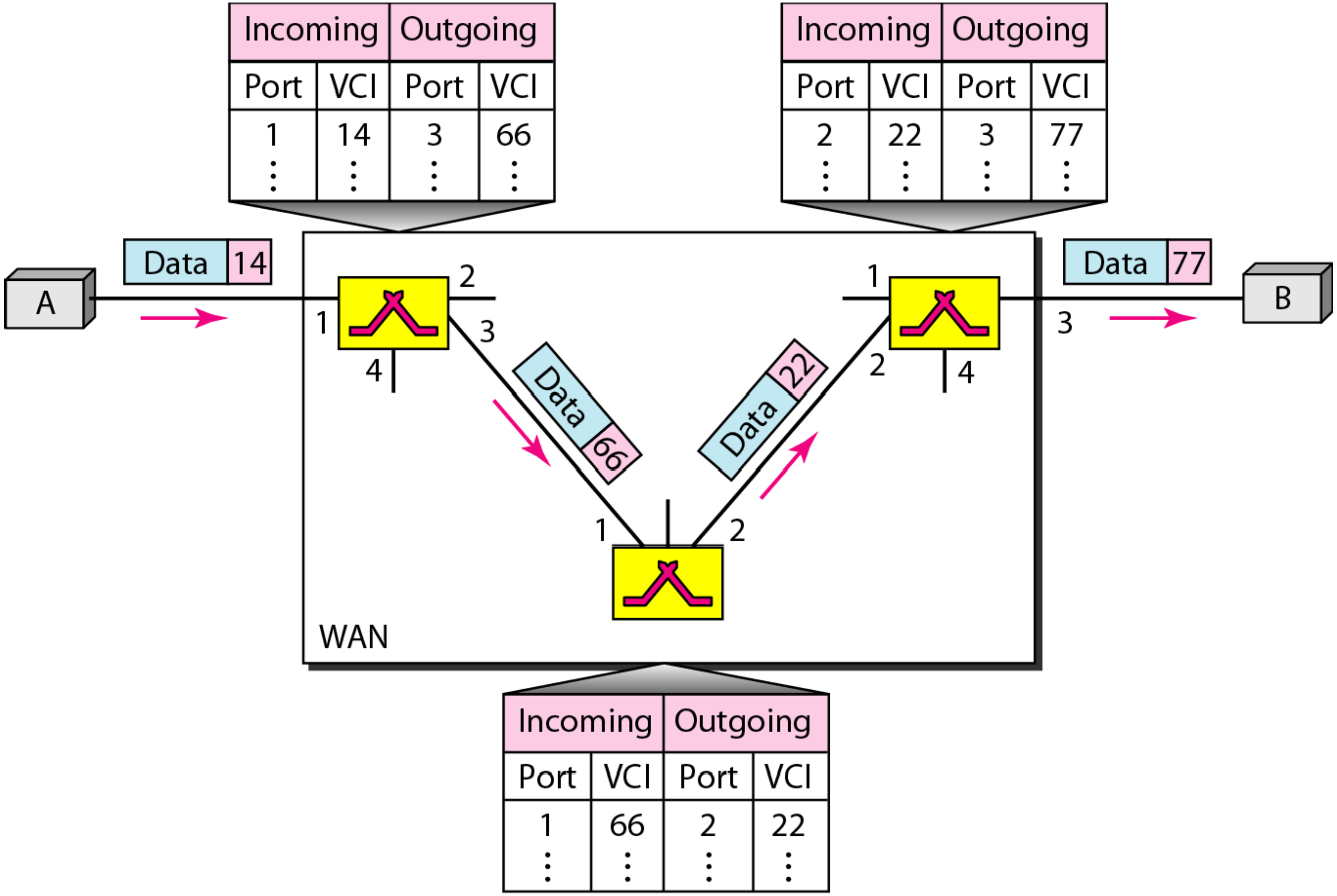


Switch and tables in a virtual-circuit network

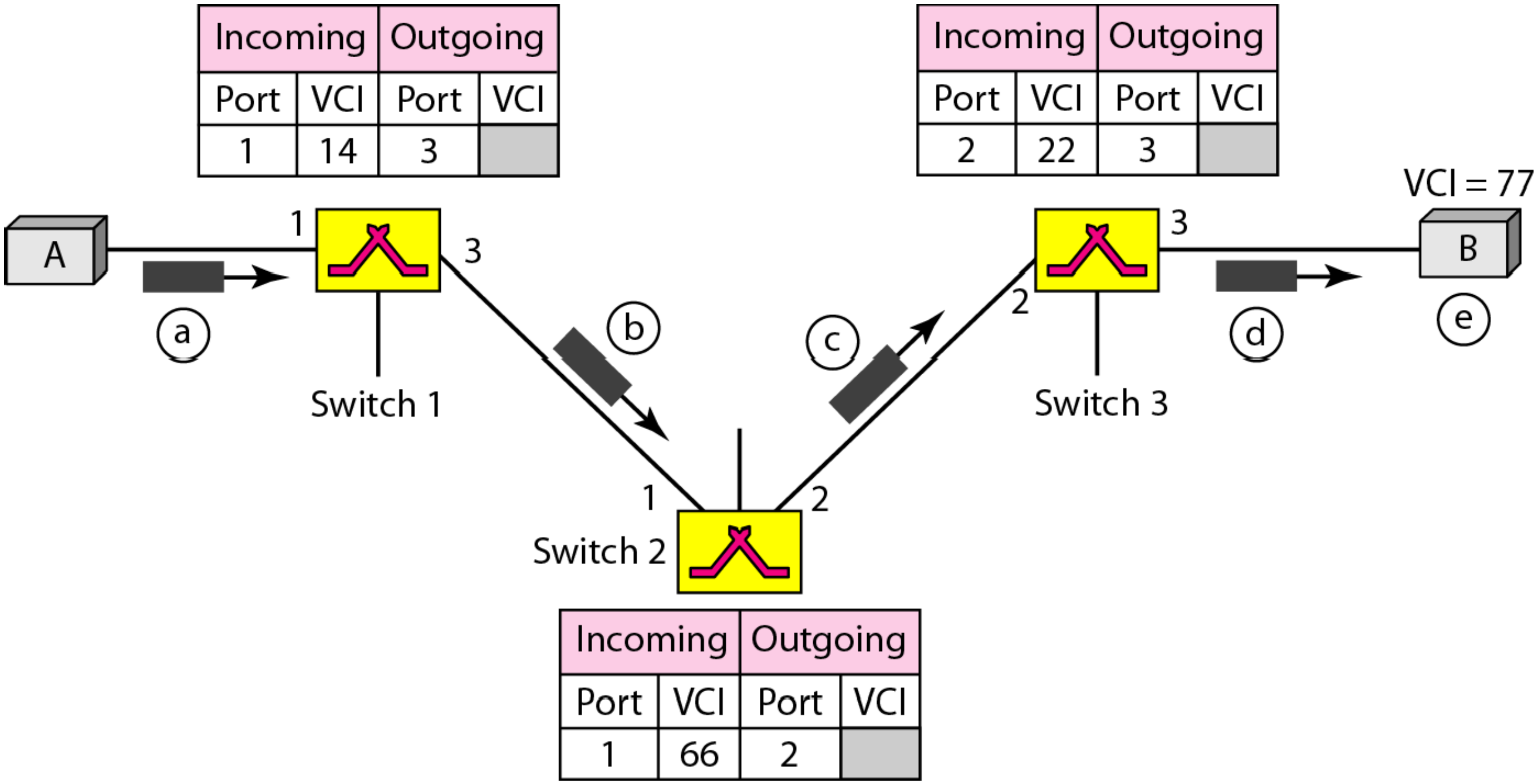
Incoming		Outgoing	
Port	VCI	Port	VCI
1	14	3	22
1	77	2	41



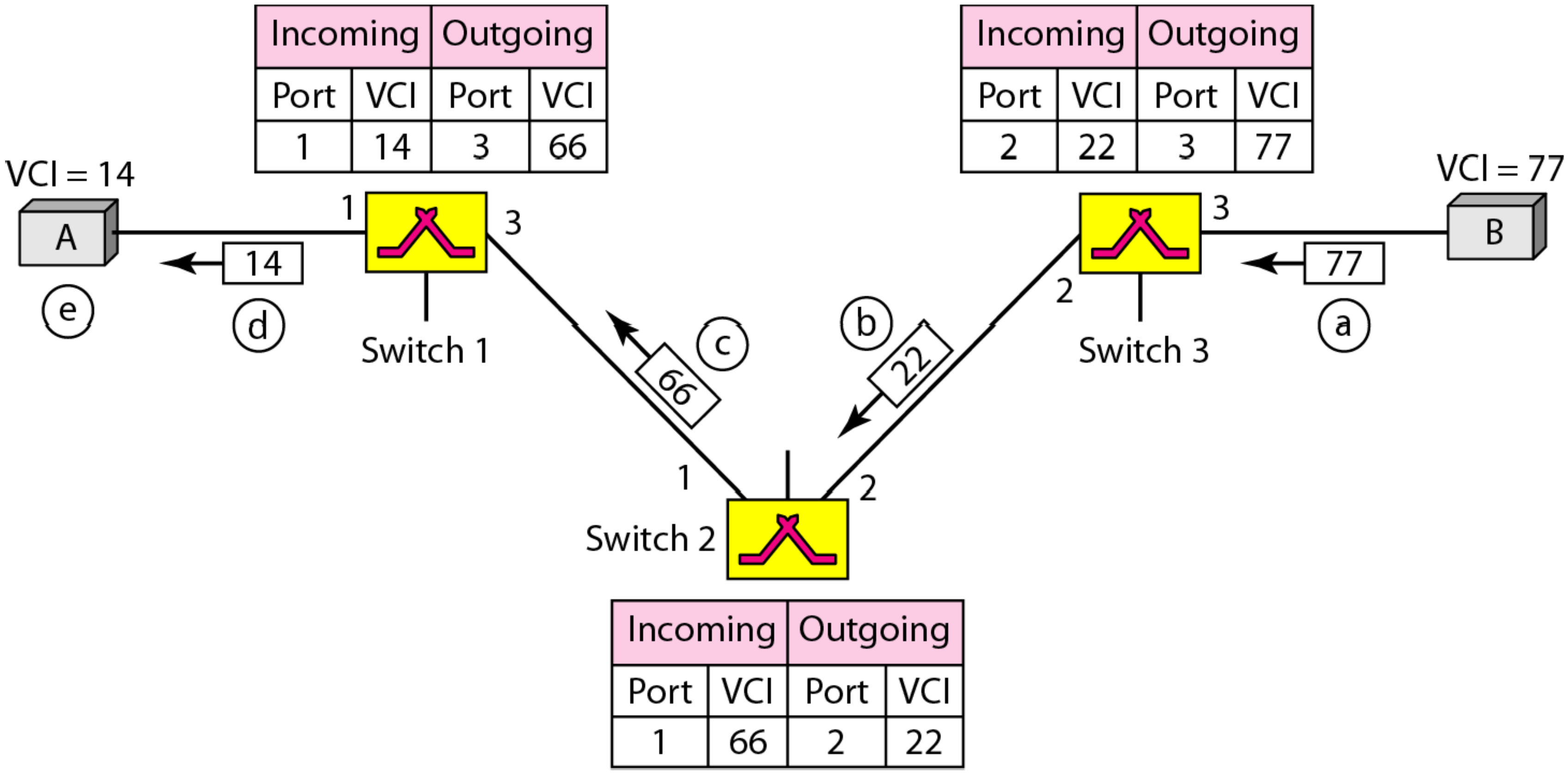
Source-to-destination data transfer in a virtual-circuit network



Setup request in a virtual-circuit network



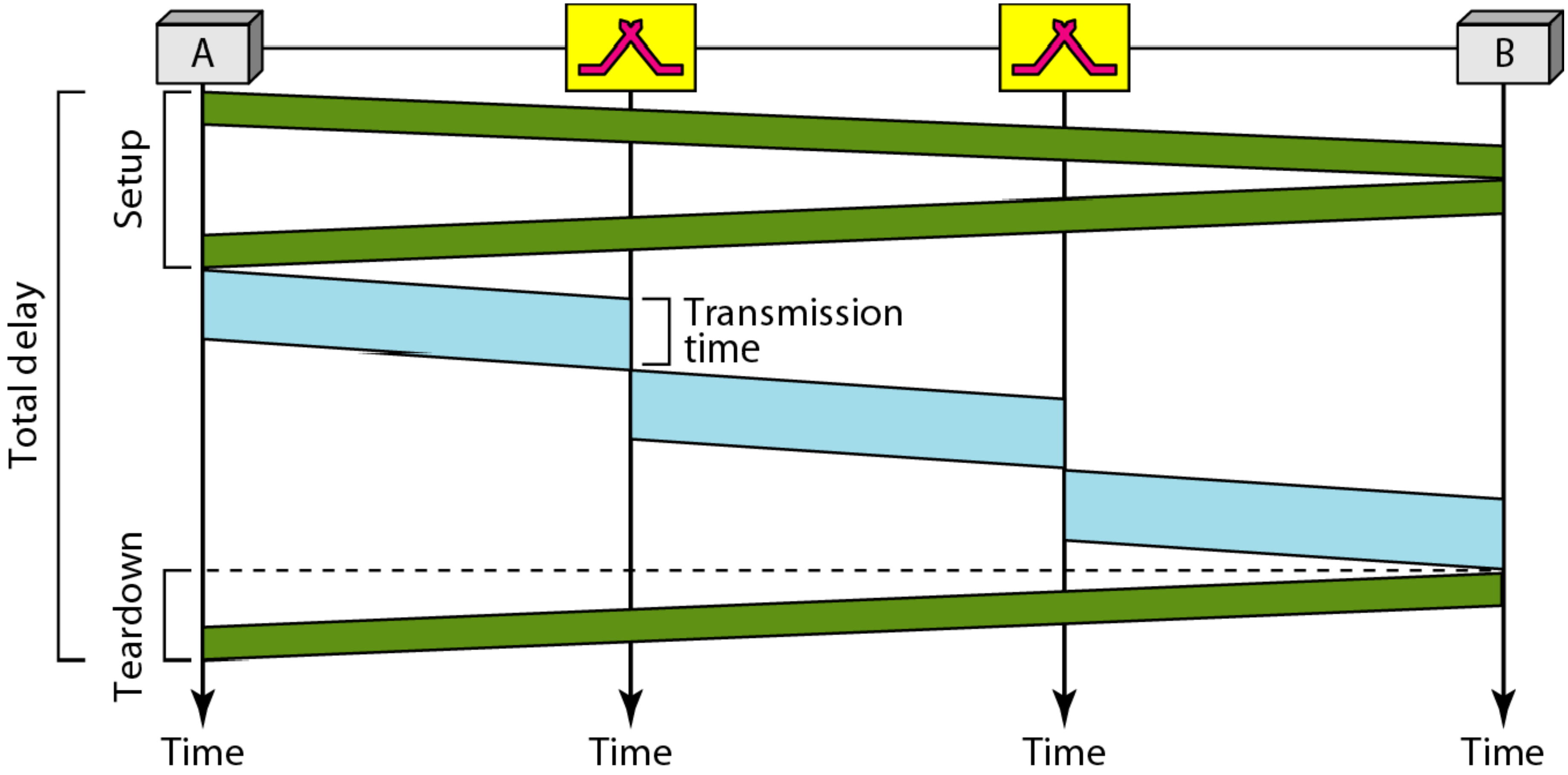
Setup acknowledgment in a virtual-circuit network



Note

In virtual-circuit switching, all packets belonging to the same source and destination travel the same path; but the packets may arrive at the destination with different delays if resource allocation is on demand.

Delay in a virtual-circuit network



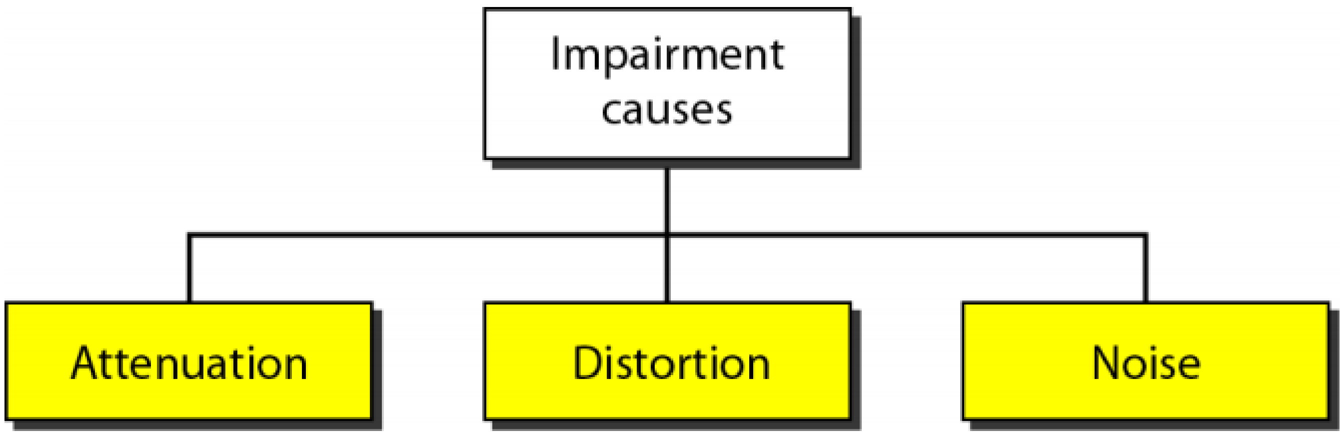
Note

Switching at the data link layer in a switched WAN is normally implemented by using

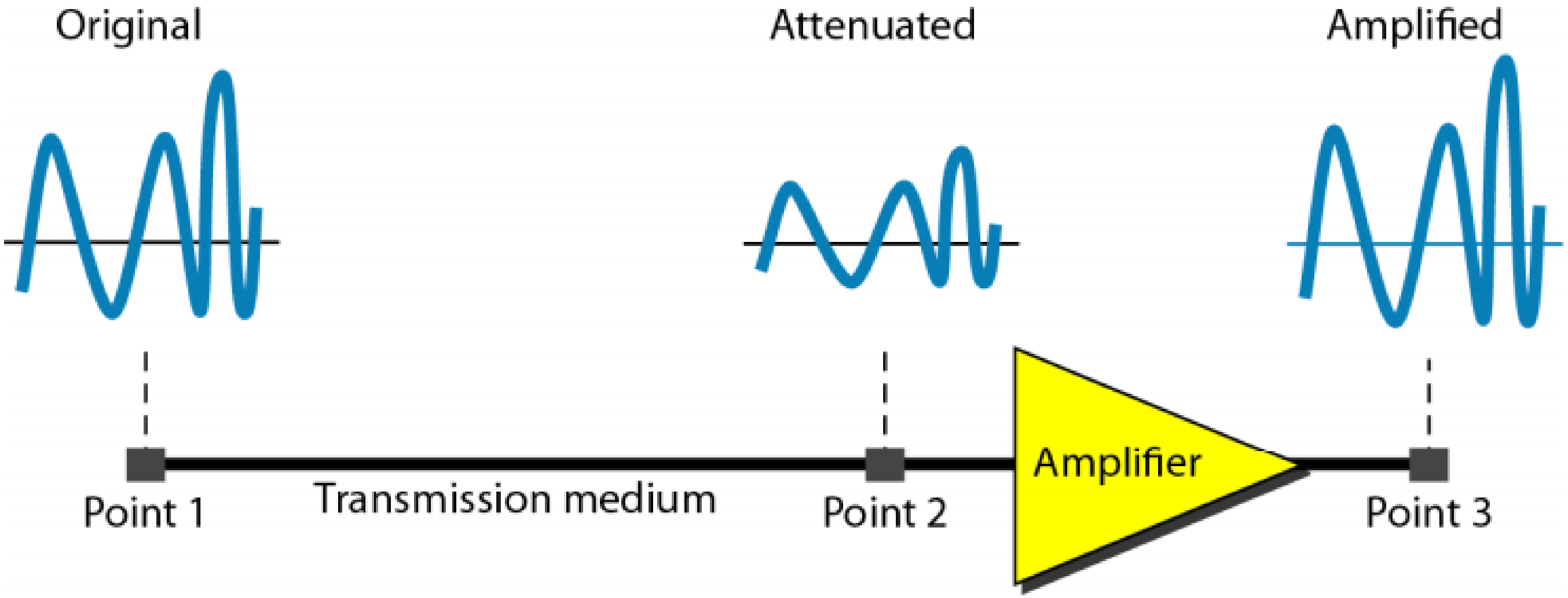
virtual-circuit techniques.

TRANSMISSION IMPAIRMENT

- Signals travel through transmission media, which are not perfect.
- The imperfection causes signal impairment.
- This means that the signal at the beginning of the medium is not the same as the signal at the end of the medium.
- What is sent is not what is received.



Attenuation



$$\mathrm{dB} = 10 \log_{10} \frac{P_2}{P_1}$$

P1 and P2 are power of signal at different points

Suppose a signal travels through a transmission medium and its power is reduced to one-half. This means that $P_2 = \frac{1}{2} P_1$. In this case, the attenuation (loss of power) can be calculated as

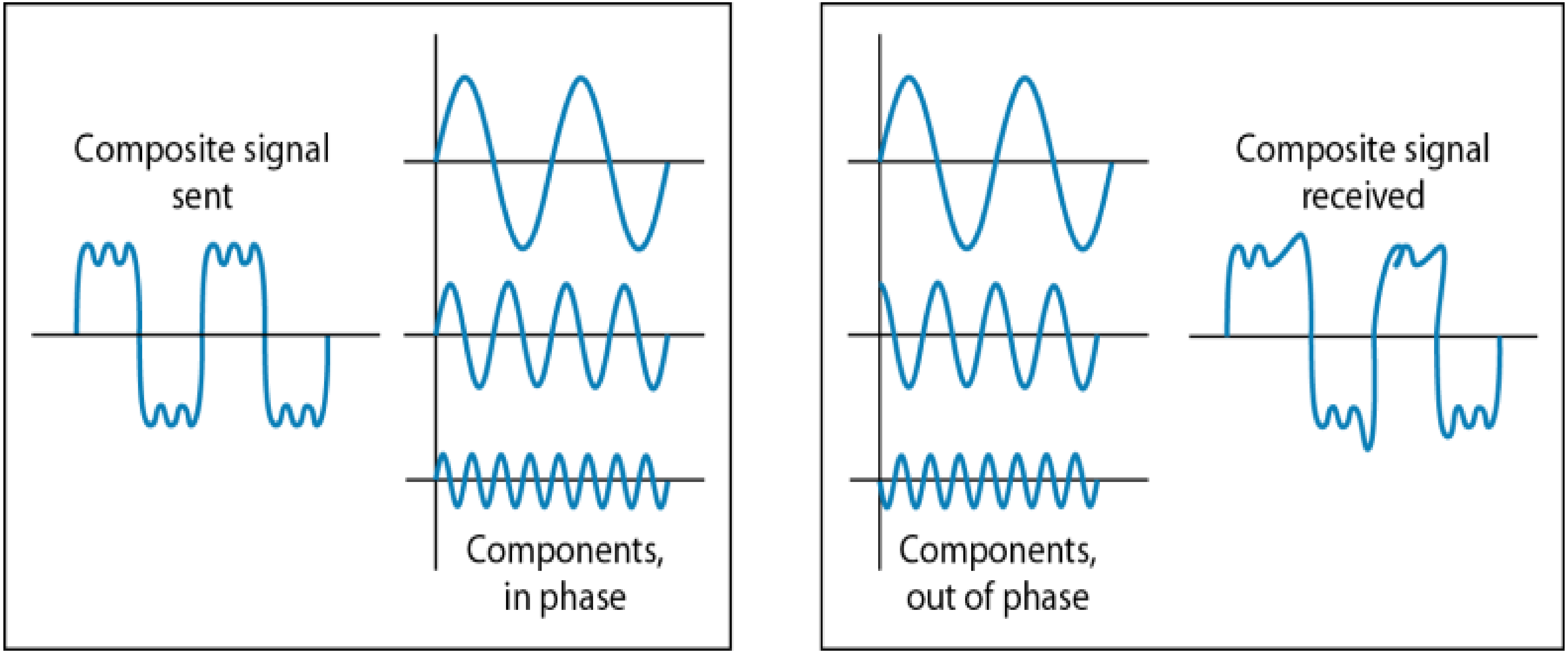
$$10 \log_{10} \frac{P_2}{P_1} = 10 \log_{10} \frac{0.5P_1}{P_1} = 10 \log_{10} 0.5 = 10(-0.3) = -3 \text{ dB}$$

A loss of 3 dB (–3 dB) is equivalent to losing one-half the power.

A signal travels through an amplifier, and its power is increased 10 times. This means that $P_2 = 10P_1$. In this case, the amplification (gain of power) can be calculated as

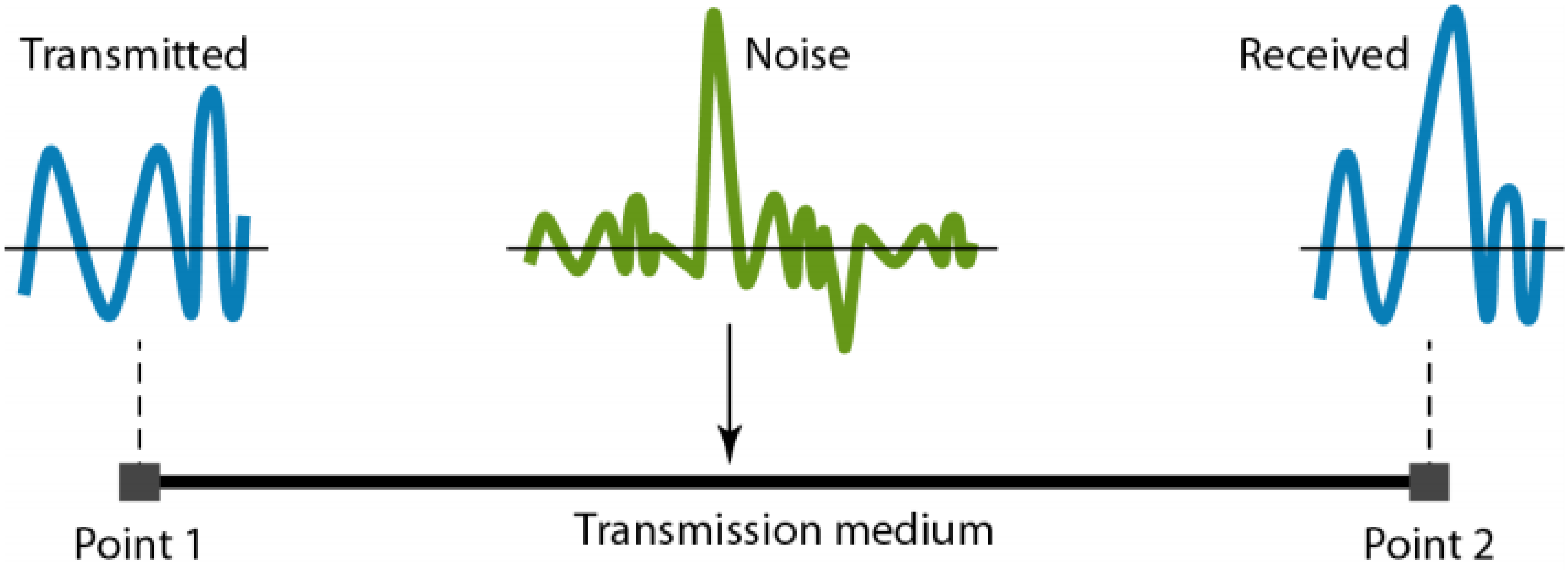
$$10 \log_{10} \frac{P_2}{P_1} = 10 \log_{10} \frac{10P_1}{P_1} = 10 \log_{10} 10 = 10(1) = 10 \text{ dB}$$

Distortion

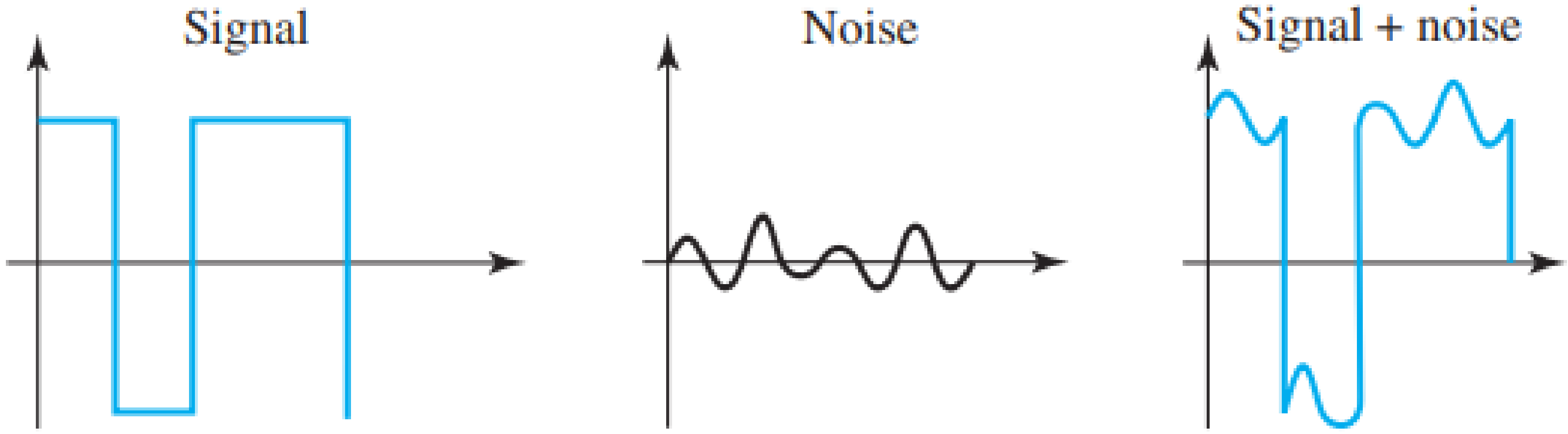


At the sender At the receiver

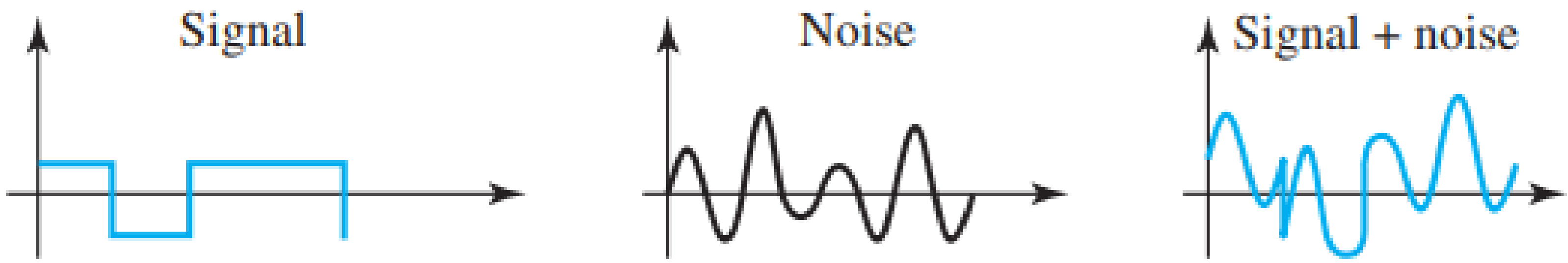
Noise



$$SNR = \frac{\text{average signal power}}{\text{average noise power}}$$



a. High SNR



b. Low SNR

Data Rate

Data rate depends on 3 factors:

1. The **bandwidth** available
2. The **level of signals** we use
3. The **quality of the channel** (the level of noise)

Two theoretical formulas were developed to calculate the data rate

- One by **Nyquist** for noiseless channel
- Another by **Shannon** for a noisy channel

Noiseless channel: Nyquist Bit rate

For a noiseless channel, the Nyquist bit rate formula defines the theoretical maximum bit rate

$$\mathbf{BitRate} = 2 \times \mathbf{bandwidth} \times \log_2 L$$

Bandwidth – Bandwidth of the channel

L – No. of signal levels used to represent data

Bit rate – Bit rate in bits per second

Increasing the level of signal may reduce the reliability of the system

Example 3.34

Consider a noiseless channel with a bandwidth of 3000 Hz transmitting a signal with two signal levels. The maximum bit rate can be calculated as

$$\textbf{BitRate} = 2 \times 3000 \times \log_2 2 = 6000 \text{ bps}$$

Example 3.35

Consider the same noiseless channel transmitting a signal with four signal levels (for each level, we send 2 bits). The maximum bit rate can be calculated as

$$\textbf{BitRate} = 2 \times 3000 \times \log_2 4 = 12{,}000 \text{ bps}$$

Example 3.36

We need to send 265 kbps over a noiseless channel with a bandwidth of 20 kHz. How many signal levels do we need?

Solution

We can use the Nyquist formula as shown:

$$265{,}000 = 2 \times 20{,}000 \times \log_2 L \longrightarrow \log_2 L = 6.625 \longrightarrow L = 2^{6.625} = 98.7 \text{ levels}$$

Noisy channel: Shannon Capacity

In reality, we cannot have noiseless channel

To determine the theoretical highest rate for noisy channel

$$\text{Capacity} = \text{bandwidth} \times \log_2(1 + \text{SNR})$$

Bandwidth – Bandwidth of channel

SNR – Signal to noise ratio

Capacity – capacity of channel in BPS

Example 3.37

Consider an extremely noisy channel in which the value of the signal-to-noise ratio is almost zero. In other words, the noise is so strong that the signal is faint. For this channel the capacity C is calculated as

$$C = B \log_2 (1 + \text{SNR}) = B \log_2(1 + 0) = B \log_2 1 = B \times 0 = 0$$

This means that the capacity of this channel is zero regardless of the bandwidth. In other words, we cannot receive any data through this channel.

Example 3.38

We can calculate the theoretical highest bit rate of a regular telephone line. A telephone line normally has a bandwidth of 3000 Hz (300 to 3300 Hz) assigned for data communications. The signal-to-noise ratio is usually 3162. For this channel the capacity is calculated as

$$C = B \log_2 (1 + \text{SNR}) = 3000 \log_2(1 + 3162) = 3000 \times 11.62 = 34{,}860 \text{ bps}$$

This means that the highest bit rate for a telephone line is 34.860 kbps. If we want to send data faster than this, we can either increase the bandwidth of the line or improve the signal-to-noise ratio.

Performance

One important issue in networking is the performance of the network—how good is it? quality of service, an overall measurement of network performance.

- **Bandwidth** - capacity of the system
  - Bandwidth in Hz
  - Bandwidth in BPS
- **Throughput** – How fast data can be send through a network
- **Latency (Delay)** – How long data take for an entire message to completely arrive at the destination from source

Latency = PT + TT + QT + PD

In networking, the term bandwidth is used in two contexts.

- The first, **bandwidth in hertz**, refers to the range of frequencies in a composite signal or **the range of frequencies that a channel can pass**.
- The second**, bandwidth in bits per second**, refers to the **speed of bit transmission in a channel** or link. Often referred to as Capacity.

Propagation speed - speed at which a bit travels though the medium from source to destination.

Transmission speed - the speed at which all the bits in a message arrive at the destination. (difference in arrival time of first and last bit)